

# Notes: Atkin, Costinot, and Fukui (RES, 2026)

## Globalization and the Ladder of Development: Pushed to the Top or Held at the Bottom?

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# 1 Big picture

- **Question.** Does international trade help countries climb the development ladder, or can trade push some countries into less capability-enhancing sectors and slow down development?
- **Core metaphor.** Countries differ in *capability*; goods differ in *complexity*. More capable countries can produce and export more complex goods. A country moves up the ladder when its capability grows.
- **Two-way relationship.** Development changes comparative advantage, but specialization can also change future development if producing complex goods generates learning-by-doing or knowledge spillovers.
- **Theory contribution.** The paper formalizes a ladder economy in which complex sectors are produced by fewer, more capable countries and in which shifting employment toward more complex sectors raises capability growth.
- **Key theoretical result.** In a pure ladder economy, trade opens markets, tougher foreign competition is stronger in simple sectors, countries shift toward their more complex sectors, and capability growth rises everywhere.
- **Mirror result.** In an inverted ladder economy, if capability-enhancing goods face tougher foreign competition, trade shifts countries away from those sectors and creates pervasive dynamic losses.
- **Empirical measurement.** The authors estimate country capability and product complexity from disaggregated Comtrade export data for 146 countries and 715 manufacturing products from 1962 to 2014.
- **Empirical identification.** They use other countries' GATT/WTO accessions as instruments for changes in a country's export complexity. WTO entry by one country changes competitive pressure in third markets and shifts the sectoral composition of other countries' exports.
- **Main empirical estimate.** Exogenous shifts toward more complex export sectors raise future capability. This supports the idea that complex sectors are capability-enhancing.
- **But the sign of trade's dynamic effect is negative.** The sectors identified as more complex tend to face *more* foreign competition, not less. Therefore trade often pushes countries away from the sectors that would raise capability growth.
- **Quantitative result.** In the calibrated model, almost every country would have higher capability under counterfactual autarky. For the median country, dynamic considerations reduce the welfare gains from trade by around 2.5 percent.
- **China application.** The rise of China produces static gains for many countries, but dynamic losses for several countries, especially African economies pushed away from their most complex sectors and toward less complex sectors.
- **Main takeaway.** International trade can generate static gains while also lowering long-run capability growth. Whether globalization pushes countries up the ladder depends on whether capability-enhancing sectors face softer or tougher foreign competition.

## 2 Data and measurement

### 2.1 Countries, products, and time period

- The empirical analysis uses product-level international trade data from the United Nations Comtrade Database.
- Products are manufacturing goods at the SITC 4-digit level.
- The baseline sample contains 715 manufacturing products.

- The country sample contains 146 countries.
- The time period runs from 1962 to 2014.
- The analysis is organized in 5-year periods, which aligns with the dynamic capability regressions.

**Interpretation:** The empirical setting is designed to capture long-run changes in product specialization rather than short-run export fluctuations. The product-level data are also essential because the concept of “complexity” is defined at a disaggregated sector level.

## 2.2 Export dummy and extensive margin

For country  $i$ , destination  $j$ , product  $k$ , and time  $t$ , the paper defines an export indicator

$$\pi_{ij,t}^k = 1\{\text{exports of product } k \text{ from } i \text{ to } j \text{ exceed } \$100,000\}, \quad (1)$$

and zero otherwise.

- The threshold is expressed in 2010 U.S. dollars.
- The focus is on the extensive margin: whether a country exports a product to a destination at all.
- The extensive margin is natural for the ladder metaphor because a country either can or cannot produce/export a sufficiently complex product.

**Interpretation:** The export indicator abstracts from export values and focuses on the set of products a country can successfully export. This is the key observable used to recover capability and complexity.

## 2.3 Capability and complexity

The paper assumes that the probability that country  $i$  exports product  $k$  to destination  $j$  increases with the interaction between country capability and product complexity:

$$\Pr(\pi_{ij,t}^k = 1) \text{ is increasing in } N_{i,t} n_t^k. \quad (2)$$

- $N_{i,t}$  is country  $i$ ’s capability at time  $t$ .
- $n_t^k$  is product  $k$ ’s complexity at time  $t$ .
- A country is more capable if it is more likely to export complex goods.
- A good is more complex if it is more likely to be exported by more capable countries.

**Interpretation:** Capability and complexity are jointly identified as a fixed point. If the United States exports a product more often than Bangladesh, and the United States is already inferred to be more capable, then that product is inferred to be more complex. Conversely, countries that export inferred complex products are classified as more capable.

## 2.4 Average complexity of a country’s product mix

The paper summarizes a country’s sectoral specialization by the average complexity of its output or export mix. A stylized version is

$$S_{i,t}^x = \sum_k \lambda_{i,k,t}^x n_t^k, \quad (3)$$

where  $\lambda_{i,k,t}^x$  is the export-based weight on product  $k$ .

- Higher  $S_{i,t}^x$  means the country specializes in more complex products.

- The law of motion for capability depends on this average complexity.
- The baseline empirical work uses export-based measures because broad product-level production data are unavailable for many countries.

**Interpretation:** The paper tests whether employment or export shifts toward more complex goods raise future capability.

## 2.5 GATT/WTO accession instruments

The main endogeneity problem is that unobserved capability shocks may change a country's specialization. The paper constructs instruments using the entry of *other countries* into GATT/WTO.

- When a country enters GATT/WTO, its trade costs with common destinations fall.
- This increases competitive pressure on other countries' exports in the same products and destinations.
- If the entrant competes with country  $i$  in complex sectors,  $i$  may be pushed away from complex goods.
- If the entrant competes with country  $i$  in simple sectors,  $i$  may be pushed toward complex goods.

The paper constructs two shifters:

- $Z_{i,t}^I$ : product-destination-level exposure to GATT/WTO entrants.
- $Z_{i,t}^{II}$ : destination-level exposure to GATT/WTO entrants.

**Interpretation:** The instruments use foreign institutional shocks to predict changes in a country's product mix. The identifying assumption is that other countries' accessions affect country  $i$ 's future capability only through induced changes in specialization, not through unobserved domestic capability shocks.

## 2.6 Baseline economy parameters

The quantitative model uses:

- $\sigma = 2.7$ , the elasticity of substitution across origins within a sector, from Broda and Weinstein.
- $\epsilon = 1.36$ , the elasticity of substitution across sectors, from Redding and Weinstein.
- $\beta = 0.279$ , the estimated dynamic spillover from average complexity to future capability.
- $\phi = 0.856$ , the persistence parameter in the capability law of motion.

**Interpretation:** The model combines estimated dynamic spillovers with external calibration of demand elasticities to compute counterfactual trade and autarky paths.

# 3 Models and empirical specifications

## 3.1 Ladder economy: preferences

Consumers have nested CES preferences. Aggregate consumption in country  $i$  at time  $t$  is

$$C_{i,t} = \left( \int (C_{i,t}^k)^{(\epsilon-1)/\epsilon} dk \right)^{\epsilon/(\epsilon-1)}, \quad (4)$$

where sector-level consumption is

$$C_{i,t}^k = \left( \sum_j (c_{j,i,t}^k)^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (5)$$

- $\epsilon > 1$  is the elasticity of substitution across sectors.
- $\sigma > 1$  is the elasticity of substitution across country varieties within sectors.
- The paper assumes  $\sigma > \epsilon$ , so within-sector varieties are more substitutable than goods from different sectors.

**Interpretation:** This assumption matters because foreign competition in a sector lowers demand for a country's variety in that sector. When trade makes competition tougher in some sectors, employment shifts away from those sectors.

### 3.2 Technology and capability threshold

Production is linear:

$$q_{ij,t}^k = A_{ij,t}^k \ell_{ij,t}^k. \quad (6)$$

In the stylized ladder economy,

$$A_{ij,t}^k = \begin{cases} A_{ij,t}, & n_t^k \leq N_{i,t}, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

- $N_{i,t}$  is country capability.
- $n_t^k$  is product complexity.
- Country  $i$  can produce product  $k$  only if its capability exceeds that product's complexity.

**Interpretation:** In the stylized theory, comparative advantage is an extensive-margin object: countries differ in the set of goods they can produce, not in productivity differences across all goods they can produce.

### 3.3 Capability accumulation

Capability evolves according to

$$\dot{N}_{i,t} = H_{i,t} (N_{i,t}, F_{i,t}^\ell), \quad (8)$$

where  $F_{i,t}^\ell$  is the cumulative distribution of employment across sectors of different complexity:

$$F_{i,t}^\ell(n) = \frac{\int_{0 \leq n_t^k \leq n} \sum_j \ell_{ij,t}^k dk}{\int \sum_j \ell_{ij,t}^k dk}. \quad (9)$$

The ladder assumption is that shifting employment toward more complex sectors raises capability growth.

**Interpretation:** This is the learning-by-doing or dynamic-spillover assumption. Firms do not internalize how their sectoral employment choices affect future national capability.

### 3.4 Trade versus autarky in the ladder economy

The autarky counterfactual shuts down trade by setting

$$(A_{ij,t}^k)' = \begin{cases} A_{ij,t}, & n_t^k \leq N_{i,t} \text{ and } i = j, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

- Under autarky, countries employ workers evenly across the goods they can produce.
- Under trade, foreign competition varies across sectors.
- If simpler sectors face tougher foreign competition, trade pushes countries toward their more complex goods.

**Interpretation:** The sign of trade’s dynamic effect depends on where foreign competition is toughest along the complexity ladder.

### 3.5 Proposition 1: ladder economy

**Proposition 1.** In a ladder economy, openness to trade raises capability and aggregate consumption at all dates in all countries.

- Foreign competition is tougher lower down the ladder.
- Countries shift employment toward the most complex sectors they can produce.
- Complex-sector employment raises capability growth.
- Dynamic gains are pervasive rather than zero-sum.

**Interpretation:** The top of the ladder does not gain at the expense of the bottom. In the pure ladder economy, trade helps every country move up.

### 3.6 Proposition 2: inverted ladder economy

**Proposition 2.** In an inverted ladder economy, openness to trade lowers capability at all dates in all countries.

- The inverted ladder reverses the key competition pattern.
- More complex, capability-enhancing sectors face tougher foreign competition.
- Trade shifts countries away from those sectors.
- Capability growth falls everywhere.

**Interpretation:** The same dynamic spillovers that create gains in the ladder economy create losses in the inverted ladder economy. The sign depends on the correlation between capability-enhancing sectors and foreign competition.

### 3.7 Quantitative productivity model

The quantitative model relaxes the strict threshold structure. Instead of requiring that only countries above a cutoff can export a product, it assumes that the probability of exporting increases with the interaction of capability and complexity. A stylized representation is

$$\pi_{ij,t}^k = \gamma_{ij,t} + \delta_{k,t} + \eta N_{i,t} n_t^k + e_{ij,t}^k. \quad (11)$$

- Country-destination-time terms absorb bilateral demand and trade costs.
- Product-time terms absorb product-specific global forces.
- The interaction  $N_{i,t} n_t^k$  captures the ladder structure.

**Interpretation:** The quantitative model allows more complex goods to be exported by many or few countries. Therefore it lets the data decide whether complex goods face softer or tougher competition.

### 3.8 Dynamic capability law

The empirical law of motion is

$$N_{i,t+\Delta} = \beta S_{i,t}^x + \phi N_{i,t} + \gamma_i + \delta_t + \varepsilon_{i,t}. \quad (12)$$

- $\Delta = 5$  years.
- $S_{i,t}^x$  is average export complexity.
- $\beta$  is the dynamic spillover parameter.
- $\phi$  captures persistence in capability.
- Country and year fixed effects absorb permanent country differences and global shocks.

**Interpretation:** A positive  $\beta$  means that shifts toward more complex goods raise future capability.

### 3.9 First-stage regression

The first-stage regression is

$$S_{i,t}^x = \alpha_1 Z_{i,t}^I + \alpha_2 Z_{i,t}^{II} + \gamma_i + \delta_t + u_{i,t}. \quad (13)$$

**Interpretation:** The instruments predict how foreign WTO entry shifts the complexity of a country's export mix through third-market competition.

### 3.10 Gains from trade decomposition

The overall gains from trade are

$$GT_{i,t} = 1 - \frac{(C_{i,t})'}{C_{i,t}}, \quad (14)$$

where  $(C_{i,t})'$  is consumption under counterfactual autarky.

The dynamic component compares trade to an autarky counterfactual in which capability evolves endogenously. The static component holds capability fixed at its trade-equilibrium level.

**Interpretation:** This decomposition separates standard static gains from trade from the dynamic effect of trade on future capability.

### 3.11 Rise of China counterfactual

The China counterfactual studies a world without China's post-1990 rise. The model compares the actual trade equilibrium to a counterfactual where China does not generate the same competitive pressure after the 1990s.

- Static effects capture cheaper imports and standard gains from trade.
- Dynamic effects capture how China's competition changes other countries' sectoral specialization and future capability.
- The focus is on 2012 as the endpoint of the main period.

**Interpretation:** China's rise can simultaneously lower prices and harm capability growth in countries pushed away from complex export sectors.

## 4 Identification and empirical design

### 4.1 Main endogeneity problem

- Countries that experience positive capability shocks may choose more complex sectors.
- Observed specialization may therefore reflect future capability rather than cause it.
- OLS estimates of the dynamic spillover  $\beta$  may be biased.



**Interpretation:** The main empirical challenge is reverse causality between capability growth and industrial structure.

## 4.2 WTO entry as an instrument

The paper uses other countries' GATT/WTO accessions to predict changes in a country's product mix.

- Entry reduces an entrant's trade costs in common destination markets.
- The entrant competes with other countries exporting similar products to those destinations.
- The induced competitive pressure shifts the sectoral composition of other countries' exports.
- These shifts vary by country because countries have different export baskets before entry.

**Interpretation:** The instrument is plausibly exogenous to country  $i$ 's unobserved capability shock because it is driven by other countries' institutional entry decisions and preexisting product-destination overlap.

## 4.3 Why two instruments?

- $Z_{i,t}^I$  uses product-destination variation and measures competition in the exact products and destinations where country  $i$  exports.
- $Z_{i,t}^{II}$  uses destination-level variation and captures broader market access effects in destinations affected by entrants.
- Using both instruments increases the strength and flexibility of the first stage.

**Interpretation:** The two instruments separate narrow product-market competition from broader destination-market exposure.

## 4.4 First-stage interpretation

Table 2 shows that the WTO-entry instruments strongly predict average export complexity. The signs of the first-stage coefficients indicate whether entrant competition pushes a country toward or away from more complex products.

**Interpretation:** The first stage is not just a statistical device. It gives the economic content of the empirical design: global entry reshapes specialization through competition.

## 4.5 Reduced-form and second-stage interpretation

The second-stage regression estimates whether exogenous increases in export complexity raise future capability. The reduced-form regression directly relates the instruments to future capability.

**Interpretation:** If the instruments shift specialization and future capability in the same direction predicted by the model, this supports dynamic spillovers from complex-sector specialization.

## 4.6 Robustness design

The paper tests the dynamic-spillover result using:

- recentered instruments;
- excluding WTO entrants;
- fixing product complexity over time;

- separate samples for poor and rich countries;
- real GDP per capita as an alternative development outcome.

**Interpretation:** The robustness checks ask whether the positive spillover estimate is an artifact of instrument construction, sample composition, time-varying product rankings, or the capability measure itself.

## 4.7 Counterfactual credibility

The counterfactual analysis combines:

- estimated complexity and capability;
- observed trade flows;
- calibrated demand elasticities;
- the estimated capability law of motion.

**Interpretation:** The counterfactuals are structural and depend on model assumptions. Their value is to map the estimated dynamic spillovers and observed competition patterns into welfare consequences.

## 5 Tables and figures

### 5.1 Figure 1: pattern of specialization after opening to trade

**Read:**

- Panel A shows the number of foreign competitors faced by a country across goods of different complexity.
- In a ladder economy, simpler goods face more competitors because more countries are capable of producing them.
- Panel B shows the employment reallocation caused by trade.
- Opening to trade shifts employment toward the most complex goods that the country can produce.

**Economic message:** This is the core theoretical mechanism: trade makes low-complexity sectors more competitive and pushes countries toward higher-complexity sectors, which can raise future capability.

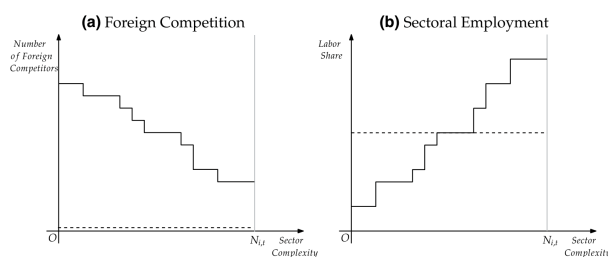


FIGURE 1

### 5.2 Figure 2 and Table 1: capability across countries and complexity across goods

**Read:**

- Figure 2 plots estimated capability for selected countries across time.
- Rich countries such as the United States and Western Europe remain high on the capability ladder.

- East Asian economies such as Korea and Thailand exhibit rapid capability growth.
- Table 1 lists the most and least complex goods. Medicaments, chemicals, cars, medical instruments, and machinery are among the most complex goods.
- Low-complexity goods include undergarments, wood panels, and plastic ornaments.

**Economic message:** The estimated measures align with intuitive notions of development: advanced countries export more complex goods, and complex goods are knowledge- and capability-intensive sectors.

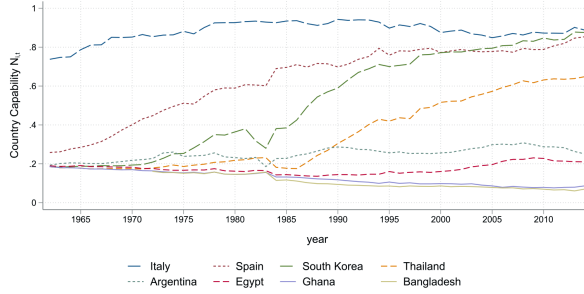


Figure 2: capability across countries

| Goods with highest $n_t^k$ (average value and rank, 1962–2014) |                                              |       |      | Goods with lowest $n_t^k$ (average value and rank, 1962–2014) |                                              |       |       |
|----------------------------------------------------------------|----------------------------------------------|-------|------|---------------------------------------------------------------|----------------------------------------------|-------|-------|
| 1                                                              | Medicaments                                  | 0.964 | 3.7  | 1                                                             | Wool undergarments                           | 0.068 | 498.1 |
| 2                                                              | Chemical Products                            | 0.874 | 11.2 | 2                                                             | Undergarments of other fibres                | 0.084 | 494.3 |
| 3                                                              | Miscellaneous non-electrical machinery parts | 0.880 | 11.8 | 3                                                             | Men's underwear                              | 0.100 | 488.1 |
| 4                                                              | Miscellaneous non-electrical machines        | 0.859 | 12.8 | 4                                                             | Wood panels                                  | 0.097 | 487.4 |
| 5                                                              | Cars                                         | 0.863 | 13.8 | 5                                                             | Aircraft tires                               | 0.090 | 486.4 |
| 6                                                              | Miscellaneous electrical machinery           | 0.830 | 15.9 | 6                                                             | Rotary converters                            | 0.083 | 485.0 |
| 7                                                              | Medical instruments                          | 0.806 | 21.9 | 7                                                             | Sheep and lamb leather                       | 0.109 | 475.7 |
| 8                                                              | Miscellaneous hand tools                     | 0.809 | 23.8 | 8                                                             | Retail yarn of more than 85% Synthetic fiber | 0.093 | 470.4 |
| 9                                                              | Electric wire                                | 0.770 | 24.2 | 9                                                             | Women's underwear                            | 0.116 | 463.2 |
| 10                                                             | Fasteners                                    | 0.760 | 26.8 | 10                                                            | Plastic ornaments                            | 0.139 | 463.8 |

Notes: This table reports the goods with the 10 highest and 10 lowest values of  $n_t^k$  from 1962 to 2014 based on each good's average rank (among goods with at least 40 years of data). The third column of each panel displays the average  $n_t^k$ , the fourth column the average rank. Complexity  $n_t^k$  is estimated year-by-year using the linear probability model described in equation (12). At every date  $t$ , complexity is assumed to be 0 for the least complex good and normalized to 1 for the most complex good. The mean of the standard deviation of each good's complexity is 0.09 (median 0.06, 90th percentile 0.23).

Table 1: complexity across goods

### 5.3 Figure 3: comparison with existing measures

Read:

- Figure 3 compares the paper's baseline measures to existing measures of capability and complexity.
- The baseline country capability measure is positively related to alternative development and complexity measures.
- The product complexity measure is also correlated with earlier product-complexity rankings.
- The comparison is made across different decades to show robustness over time.

**Economic message:** The new measures are not arbitrary. They are consistent with prior empirical approaches while being designed for the paper's theoretical ladder structure.

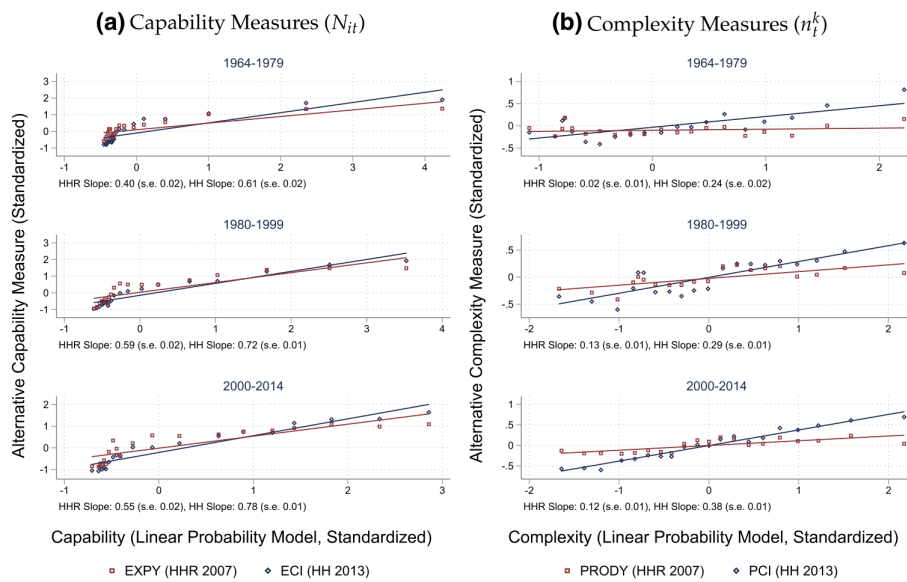


Figure 3

## 5.4 Figure 4: WTO-entry instruments

Read:

- Figure 4 plots the time path of the WTO-entry instrument for selected countries.
- China's WTO entry in 2001 creates visible jumps in exposure for several countries.
- For countries whose exports overlap with China in complex products, the instrument implies pressure away from complex sectors.
- For countries competing with entrants in less complex sectors, the instrument can push the export mix toward more complex goods.

**Economic message:** WTO entry provides the exogenous variation used to identify how changes in sectoral complexity affect future capability.

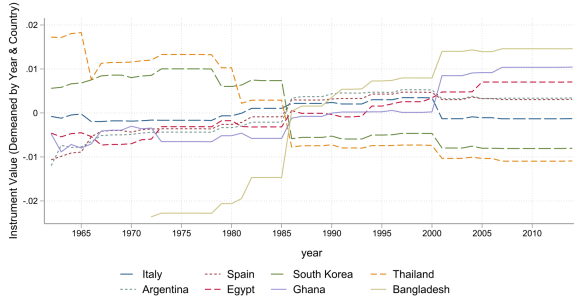


Figure 4

|                                | Average complexity $S_{i,t}^j$ |                      | Country Capability $N_{i,t+\Delta}$ |                           |                                         |                                         |
|--------------------------------|--------------------------------|----------------------|-------------------------------------|---------------------------|-----------------------------------------|-----------------------------------------|
|                                | (1)<br>FS                      | (2)<br>FS            | (3)<br>OLS                          | (4)<br>IV ( $Z_{i,t}^j$ ) | (5)<br>IV ( $Z_{i,t}^j, Z_{i,t}^{j*}$ ) | (6)<br>RF ( $Z_{i,t}^j, Z_{i,t}^{j*}$ ) |
| WTO entrant IV $Z_{i,t}^j$     | -0.647***<br>(0.214)           | -0.151<br>(0.224)    |                                     |                           |                                         | -0.154***<br>(0.0511)                   |
| WTO entrant IV $Z_{i,t}^{j*}$  |                                | -4.075***<br>(0.779) |                                     |                           |                                         | -0.624***<br>(0.219)                    |
| Average complexity $S_{i,t}^j$ |                                |                      | 0.00002**<br>(0.00389)              | 0.360**<br>(0.147)        | 0.279***<br>(0.0676)                    |                                         |
| Initial capability $N_{i,t}$   |                                |                      | 0.934***<br>(0.0208)                | 0.831***<br>(0.0477)      | 0.856***<br>(0.0353)                    | 0.932***<br>(0.0209)                    |
| Country and year FEs           | Yes                            | Yes                  | Yes                                 | Yes                       | Yes                                     | Yes                                     |
| Observations                   | 7,617                          | 7,617                | 6,872                               | 6,872                     | 6,872                                   | 6,872                                   |
| $R^2$                          | 0.586                          | 0.592                | 0.987                               | 0.622                     | 0.707                                   | 0.987                                   |
| Clusters                       | 1,588                          | 1,588                | 1,438                               | 1,438                     | 1,438                                   | 1,438                                   |
| CD F-Stat                      |                                |                      |                                     | 30.15                     | 37.25                                   |                                         |
| KP F-Stat                      |                                |                      |                                     | 8.538                     | 8.767                                   |                                         |

Notes: Columns 1 and 2 of this table report estimates of  $\alpha_1$  and  $\alpha_2$  in equation (17) by regressing the average complexity of country  $i$ 's export mix in year  $t$ ,  $S_{i,t}^j$ , on our two WTO entry instruments,  $Z_{i,t}^j$  and  $Z_{i,t}^{j*}$ , as well as country and year fixed effects. Instruments defined in equations (13) and (14) constructed from data on WTO entry events and the complexity-weighted overlap between  $i$  and the WTO entrant's export mixes 5-years prior to entry. Measures of complexity  $S_{i,t}^j$  calculated using the linear probability model estimation of equation (12). Columns 3-6 report estimates of  $\beta$  and  $\phi$  in equation (14) by regressing capability of country  $i$  in  $t + 5$ ,  $N_{i,t+\Delta}$ , with  $\Delta = 5$ , on the average complexity of  $i$ 's export mix,  $S_{i,t}^j$ , initial capability,  $N_{i,t}$ , and country and year fixed effects, using the baseline measures of complexity  $S_{i,t}^j$  and capability  $N_{i,t}$  from the same linear probability model. Columns 4 and 5 instrument average complexity of country  $i$ 's export mix,  $S_{i,t}^j$ , by the two WTO entry instruments  $Z_{i,t}^j$  and  $Z_{i,t}^{j*}$ . Column 6 reports the reduced form regression corresponding to Column 5. Standard errors clustered at the 5-year-period-country level. \*, \*\*, and \*\*\* denote statistical significance levels of 10%, 5% and 1%, respectively.

## 5.5 Table 2: changes in capability and industrial structure

Read:

- Columns 1 and 2 show the first-stage relationship between the WTO-entry instruments and average export complexity.
- Columns 3-6 estimate the capability law of motion.
- OLS gives a small positive coefficient on average complexity.
- IV estimates are much larger and statistically significant.
- The preferred two-instrument IV estimate is  $\beta = 0.279$  and  $\phi = 0.856$ .
- The reduced-form column confirms that the instruments predict future capability.

**Economic message:** Exogenous shifts toward more complex sectors raise future capability. This is the paper's main evidence for dynamic spillovers from complex-sector specialization.

## 5.6 Table 3: robustness of dynamic spillovers

Read:

- The baseline IV estimate remains positive and significant across robustness checks.
- Recentering the instruments does not eliminate the effect.
- Excluding WTO entrants still yields a positive and significant coefficient.
- Fixing complexity over time produces a similar positive estimate.

- The effect is positive for both poor and rich countries.
- Using log GDP per capita instead of capability also gives a positive relationship.

**Economic message:** The positive spillover from complex-sector specialization is robust. The dynamic losses from trade later in the paper are not due to a negative spillover estimate; they arise because trade pushes countries away from these sectors.

TABLE 3  
Changes in capability and industrial structure: robustness checks

|                                  | Country capability $N_{i,t+\Delta}$ |                       |                           |                                    |                          | $\ln GDP_{i,t+\Delta}$   |                      |
|----------------------------------|-------------------------------------|-----------------------|---------------------------|------------------------------------|--------------------------|--------------------------|----------------------|
|                                  | (1)<br>Baseline                     | (2)<br>Recenter<br>IV | (3)<br>No WTO<br>Entrants | (4)<br>Fixed $\alpha_i^e$<br>in IV | (5)<br>Poor<br>Countries | (6)<br>Rich<br>Countries | (7)                  |
| Average complexity $S_{i,t}^e$   | 0.279***<br>(0.0870)                | 0.311***<br>(0.105)   | 0.466**<br>(0.200)        | 0.307**<br>(0.153)                 | 0.21***<br>(0.0908)      | 0.259**<br>(0.0952)      | 0.905**<br>(0.409)   |
| Initial capability $N_{i,t}$     | 0.856***<br>(0.0353)                | 0.847***<br>(0.0397)  | 0.746***<br>(0.0584)      | 0.848***<br>(0.0499)               | 1.073***<br>(0.0458)     | 0.780***<br>(0.0427)     |                      |
| $\ln GDP$ per capita $GDP_{i,t}$ |                                     |                       |                           |                                    |                          |                          | 0.758***<br>(0.0327) |
| Country and year FEs             | Yes                                 | Yes                   | Yes                       | Yes                                | Yes                      | Yes                      | Yes                  |
| Observations                     | 6,872                               | 6,872                 | 2,806                     | 6,872                              | 2,812                    | 2,801                    | 6,107                |
| $R^2$                            | 0.307                               | 0.679                 | 0.545                     | 0.682                              | 0.743                    | 0.670                    | 0.599                |
| Clusters                         | 1,438                               | 1,438                 | 591                       | 1,438                              | 595                      | 586                      | 1,269                |
| CD F-Stat                        | 37.25                               | 28.61                 | 11.98                     | 9.552                              | 15.01                    | 55.77                    | 64.98                |
| KP F-Stat                        | 8.767                               | 6.955                 | 4.500                     | 2.371                              | 6.277                    | 7.815                    | 17.11                |

Notes: This table reports estimates of  $\beta$  and  $\phi$  in equation (14) by regressing capability of country  $i$  in  $t + 5$ ,  $N_{i,t+\Delta}$ , with  $\Delta = 5$ , on the average complexity of  $i$ 's export mix,  $S_{i,t}^e$ , initial capability,  $N_{i,t}$ , and country and year fixed effects, using the baseline measures of complexity  $\alpha_i^e$  and capability  $N_{i,t}$  from the linear probability model estimation of equation (12). All columns use the two-instrument IV strategy to instrument for  $S_{i,t}^e$ . Column 1 reports our baseline estimates (Column 5 of Table 2). Column 2 re-centers both IVs by drawing from a DGP for WTO entry 1000 times, as described in footnote 20, and subtracting off the mean value of the 1000 draws for each  $i$  pair. Column 3 re-estimates  $N_{i,t}$  and  $\alpha_i^e$  using only countries who do not enter the WTO during our sample period, 1962–2014, and also excludes these countries from the regression sample (the instrument is still formed by exploiting the many WTO entry events and the export overlap of the non-entrants sample with the WTO entrants). Column 4 constructs the instruments using a fixed value of complexity, the value of  $\alpha_i^e$  in the first year product  $i$  appears, to weight the export overlap with WTO entrants. Columns 5 and 6 split the sample by poor and rich countries, defined as whether the country-level mean of log real GDP per capita across all years (using RGDPiE from the Penn World Tables 9.0, Feenstra et al., 2015) is above or below the sample median for that object. The median split is identical if log real GDP per capita is first demeaned by year before taking country means. Column 7 replaces the dependent variable capability by log GDP per capita (again using RGDPiE from the Penn World Tables 9.0). Standard errors clustered at the 5-year period-country level. \*, \*\*, and \*\*\* denote statistical significance levels of 10%, 5% and 1%, respectively.

TABLE 4  
Baseline economy

| Parameter                       | Value | Choice calibration                    |
|---------------------------------|-------|---------------------------------------|
| Panel A: Nested CES preferences |       |                                       |
| $\sigma$                        | 2.7   | Broda and Weinstein (2006)            |
| $\epsilon$                      | 1.36  | Redding and Weinstein (2024)          |
| Panel B: Dynamic spillovers     |       |                                       |
| $\beta$                         | 0.279 | Baseline estimate (Table 2, Column 3) |
| $\phi$                          | 0.856 | Baseline estimate (Table 2, Column 3) |

## 5.7 Table 4: baseline economy calibration

Read:

- The nested-CES preference parameters are  $\sigma = 2.7$  and  $\epsilon = 1.36$ .
- The dynamic-spillover parameters are  $\beta = 0.279$  and  $\phi = 0.856$ .
- These parameters combine external demand estimates with the paper's estimated capability law of motion.

**Economic message:** The counterfactual results are disciplined by both trade elasticities and the estimated dynamic effect of average complexity.

## 5.8 Figure 5: welfare consequences of international trade

Read:

- Panel A shows static gains from trade across countries in 2012.
- Static gains are positive for almost all countries.
- Panel B shows dynamic gains from trade.
- Dynamic gains are negative for almost all countries.
- For the median country, dynamic considerations reduce gains from trade by about 2.5 percent.

**Economic message:** The paper does not deny static gains from trade. Its contribution is to show that the dynamic specialization channel can offset part of those gains.

## 5.9 Figure 6: complex goods face tougher competition

Read:

- Figure 6 plots measures of foreign competition against product complexity.
- The empirical relationship is the opposite of the pure ladder economy.
- More complex goods face tougher foreign competition.

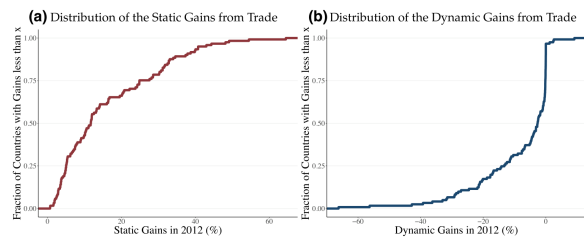


FIGURE 5

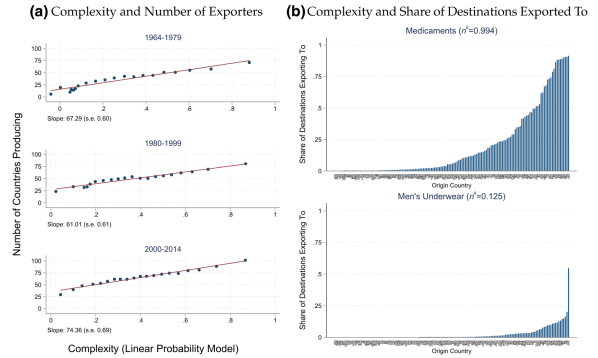


FIGURE 6

- This explains why trade tends to lower capability in the quantitative counterfactual.

**Economic message:** The reason dynamic losses are pervasive is not that complex sectors fail to raise capability. It is that trade makes those sectors harder to specialize in.

## 5.10 Figure 7: welfare consequences from the rise of China

Read:

- Figure 7 decomposes the static and dynamic consequences of China's rise.
- Static effects are generally positive because China's emergence lowers prices and expands trade opportunities.
- Dynamic effects are negative for some countries, especially countries pushed away from their more complex sectors.
- African countries are disproportionately represented among those with large dynamic losses.

**Economic message:** China's rise illustrates the paper's main mechanism in a concrete historical episode: trade can create immediate gains while slowing capability growth in countries whose complex sectors are displaced.

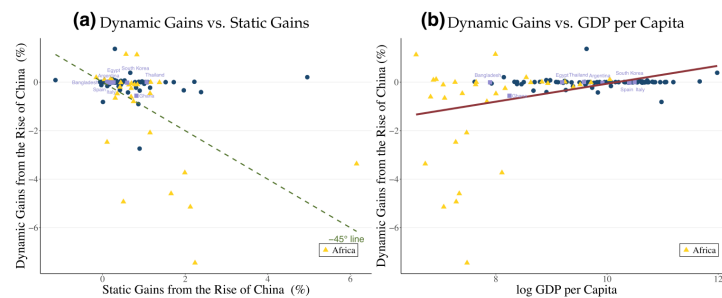


FIGURE 7

## 6 Short summary

- The paper formalizes a development ladder in which countries differ in capability and goods differ in complexity.
- In a ladder economy, trade pushes countries toward more complex goods and raises capability growth everywhere.

- In an inverted ladder economy, trade pushes countries away from capability-enhancing sectors and lowers capability growth everywhere.
- The authors estimate capability and complexity from Comtrade export data for 146 countries and 715 manufacturing products from 1962 to 2014.
- Rich countries are estimated to be more capable, and complex goods include medicines, chemicals, cars, and machinery.
- GATT/WTO accessions of other countries provide instruments for changes in a country's export complexity.
- IV estimates show that specialization in more complex goods raises future capability and real GDP per capita.
- Quantitative counterfactuals show that the sectors that raise capability tend to face tougher foreign competition.
- As a result, trade generates pervasive dynamic capability losses, though static gains from trade remain positive.
- The rise of China creates static gains for many countries but dynamic losses for some African countries that are displaced from complex sectors.

## 7 Conclusion

### 7.1 Core conclusion

Atkin, Costinot, and Fukui show that trade and development are linked through specialization. If complex sectors generate capability growth and trade shifts countries toward those sectors, globalization pushes countries up the ladder. But if those sectors face tougher foreign competition, trade can hold countries at the bottom by pushing them away from capability-enhancing activities.

### 7.2 Economic intuition

The economic logic has two parts. First, producing complex goods builds capability. Second, trade changes which goods countries produce. Whether trade improves development therefore depends on the interaction between sectoral learning and the pattern of foreign competition. In the data, the first force is positive: complex goods raise capability. The second force is adverse: complex goods face tougher competition. The combination produces dynamic losses from trade.

### 7.3 Takeaway

The paper reframes the globalization-development debate. The question is not only whether trade creates static gains. It is whether trade reallocates employment toward sectors that build future capabilities. In the calibrated model, the answer is often no: globalization creates static gains but can weaken the long-run accumulation of productive capability.